Class #18 - Determinants

1. Goal: To show that det A determines whether a square matrix A is __________or not.

2. Note:

   (a) det A is defined only for an __________matrix A.

   (b) A is invertible __________det $A \neq 0$.

   (c) Notation: det $A = |A|$

3. Computation of Determinants

   Example: Find the determinant of the matrix $A = \begin{bmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{bmatrix}$.

   Pick a row or column which is easy to work with.

   $$\det A = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}|A_{ij}| = \sum_{i=1}^{n} a_{ij}(-1)^{i+j}|A_{ij}| \qquad (*)$$

   This begins by defining the determinant of a 1×1 matrix to be the value of the single entry.

   We call the __________$(-1)^{i+j}|A_{ij}|$ the $(i, j)-$cofactor and (*) is called the *cofactor expansion.*

$\det A =$

So, A is __________.

4. Fact: $\det A$ is __________.

5. Theorem: $\det A = \det$ __________.

6. Determinant of a triangular matrix:

Theorem: det of a triangular matrix = __________.

7. Short-cut for 3×3 matrix:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} =$$

From 4×4 and onward, we do cofactor expansion.